Nirav Kanthed

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Education

University of California at San Diego — La Jolla, CA

Bachelor of Science in Data Science | GPA: 3.704/4.0 — *Expected Graduation: June 2028*

- **Courses:** Data Structures & Algorithms, Machine Learning, Statistics & Applications, Multivariable Calculus
- **Leadership & Clubs: Director** @ Association for Computing Machinery, **President** of Data Hacks Team @ DS3

Experience

Amazon - *Prime Video Team* — Seattle, WA

Software Development Engineering Intern — *Sep 2026 - Dec 2026*

- Saved engineers **5 hours/week** of manual review by building a ML workflow agent that configures and monitors engagement and monetization model training jobs, explains errors, and verifies logs, file paths through a dry-run gate for the end-to-end ML pipeline.
- Developed a neural network engagement model for Prime Video A/B experiments and evaluated it using composite metrics.
- Automated model explainability reporting across **78+ statistics** by building an automated workflow that retrieves trained recommendation models and generates reports from selected metrics, reducing time spent running evaluation scripts by hand.

IBM - *HashiCorp Team* — Austin, TX

Software Development Engineering CO-OP — *Aug 2026 - Sep 2026*

- Detected new API documentation gaps using 15 custom vacuum rules by building Go tooling in GitHub Actions to compare OpenAPI specifications/documentation before and after code changes and testing the rules with 30 passing and failing test cases.
- Enabled one-click OpenAPI documentation fixes by integrating a Claude agent into GitHub Actions to analyze Go endpoints and vacuum findings, then propose missing path descriptions, operation summaries, and typed response schemas in pull requests.
- Validated OpenAPI fixes in isolated builds by regenerating specs and checking Swagger UI and TypeScript client compatibility.

Lemn-Lab - *Language Learning Research* — La Jolla, CA

Machine Learning Researcher — *Jan 2025 - Jan 2026*

- Reduced word error rate by 52% relative to baseline across 7 underrepresented languages by fine-tuning OpenAI Whisper with PyTorch on Google FLEURS & other public speech datasets, using Bayesian hyperparameter sweeps to compare training configs.
- Supported language-specific tokenization across 7 underrepresented languages by contributing to AdaptBPE tokenizer development for Whisper, training tokenizer variants and evaluating transcription results during model fine-tuning.
- Loaded data onto Kubernetes PVCs, trained an AdaptBPE tokenizer, fine-tuned Whisper, and logged metrics in Weights & Biases.

SanQuest Inc - *Provides Cloud Solutions* — Irving, TX

Software Development Engineer Intern — *Jun 2025 - Sep 2025*

- Built an HR onboarding dashboard with React, FastAPI, and PostgreSQL to centralize employee documents and track missing, submitted, and verified records, giving HR staff one place to see progress and identify incomplete paperwork.
- Developed FastAPI endpoints for document uploads, retrieval, and status tracking, integrated the DocuSign API for onboarding workflows, stored files in AWS S3 with signed URLs, and saved document metadata in PostgreSQL.
- Automated database migrations and integration tests with GitHub Actions to verify document uploads before deployment.

Projects

TokenLens | *TypeScript, scikit-learn, LangGraph, MongoDB* — Aug 2026 - Present

- Built a Claude Code plugin and Cursor/VS Code extension that predict how many tokens a coding response will use, achieving a 180-token median absolute prediction error on Claude Haiku 4.5 with a scikit-learn model trained on 12,052 coding prompts.
- Ranked prompt rewrites by predicted token savings using a LangGraph pipeline that gates generation using acceptance/rejection history in MongoDB Atlas, rescores candidates with the prediction model, and exposes the optimizer through an MCP server.

JobTrackr | *React, Spring Boot, PostgreSQL, AWS* — Jun 2025 - Present

- Built an internship tracking platform supporting ~250 users with React, Spring Boot, and PostgreSQL, implementing JWT authentication, application search and filters, and CSV imports; deployed it on AWS EC2.
- Delivered targeted feedback on ~70 resumes by building a TF-IDF and cosine similarity matching pipeline that compares resumes with job descriptions, scores relevance, and identifies missing keywords.

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, Golang, HTML/CSS
Frameworks & Tools: PyTorch, TensorFlow, Pandas, NumPy, scikit-learn, Whisper, NLTK, spaCy, React, Spring Boot, Selenium, PostgreSQL, Kubernetes, AWS (EC2, S3, RDS), Docker, Git, GitHub, GitHub Actions, Weights & Biases, Tableau, Jupyter, REST APIs